2.6 Domain Randomization and Latent Adaptation

2.6.1 Robustness Through Randomized Training

A model trained on one simulated plant inherits its parameters as assumptions. Domain randomization instead trains across a family, first for visual appearance [Tobin 17] and then for dynamics such as mass, friction, gain and latency [Peng 18]. The hydraulic family in Chapter 4 varies bulk modulus, valve gain, leakage, friction and payload, producing robustness to physically plausible plant variation.

2.6.2 From Robustness to Adaptation

Figure 2.4 contrasts a policy trained for one robust response across the plant family with an adaptive stack that conditions its response on the current plant.

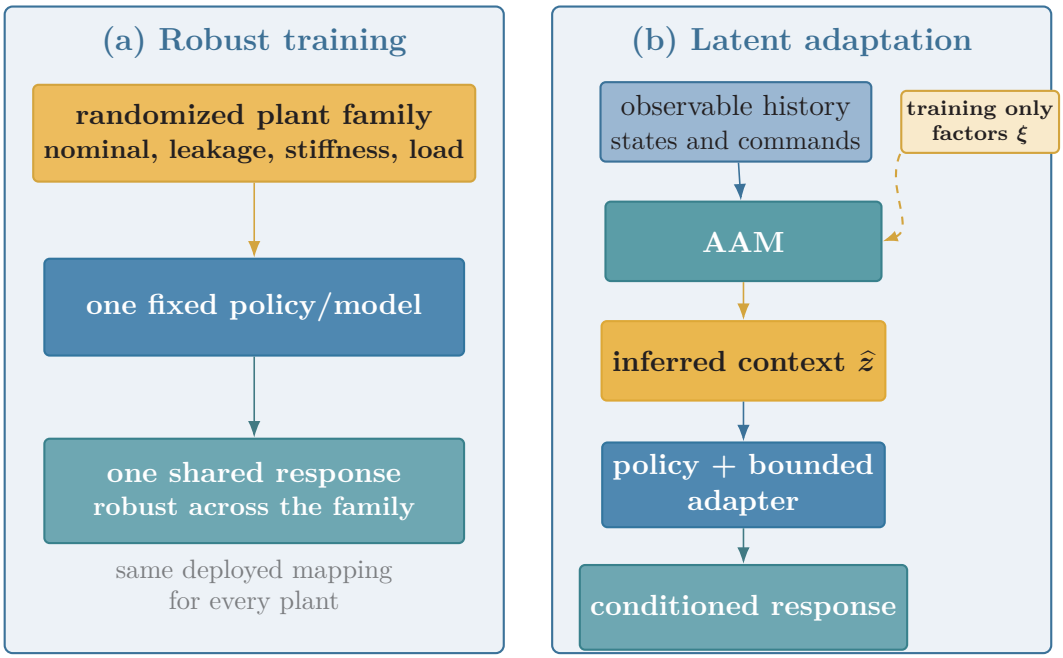


Figure 2.4: Comparison of robust training and latent adaptation. Robust training applies one fixed response across the randomized plant family. The adaptive stack uses privileged factors only to supervise training; during deployment, the AAM infers context from observable history and conditions the downstream response.

Teacher–student methods encode known simulation factors into a privileged latent and train an estimator to recover it from observable history. Lee et al. [Lee 20] used this construction for hidden terrain properties; Kumar et al. [Kumar 21] developed rapid motor adaptation. Unlike classical online parameter estimation [Åström 95], the learned latent need not have a direct physical interpretation. It can also condition a dynamics model rather than a policy, as in the contextual neural ODE of Yu et al. [Yu 26].

The AAM in Chapter 5 estimates a two-dimensional latent from one second of causal observable history. In AIF, that estimate enters the actor observation and a bounded adapter around frozen inverse dynamics; it does not condition the ROM, forward model